Forming and solving differential equations [Cambridge revision]

An equation about a rate — and the whole of the derivative and the integral put to work.

LESSON 97–100 | 4 × 40 MIN | ALGEBRA 11, QO'SHIMCHA BO'LIM | P2/P3 10.1–10.4

LESSON CLOCK

25' | 30' | 40' | 35' | 25' | 5

What a differential equation says	25 min
Forming one	30 min
Separating the variables	40 min
Particular solutions	35 min
Interpreting	25 min
Homework	5 min

LEARNING OBJECTIVES

- Form a differential equation from a statement about a rate of change.
- Solve a separable equation, and find the general solution.
- Use an initial condition to find the particular solution.
- Interpret the solution, including its long-term behaviour.

TEXTBOOK

National	pp. 441–460
Cambridge	Pure Mathematics 2 & 3, pp. 224–245
Workbook	Exercise 10A–10D

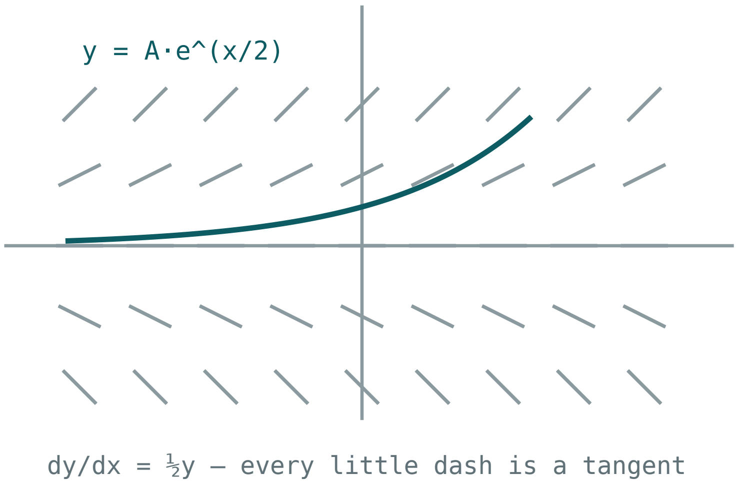
01 Explanation

What a differential equation says

An ordinary equation is a statement about a **number**. A differential equation is a statement about a **rate**, and its solution is a **function**.

$$\frac{dy}{dx} = 2x \text{ has solution } y = x^2 + c$$

Note the c: there is a whole family of solutions, one curve through each point of the plane. A single extra condition — a point the curve passes through — picks one out.



Every dash is the gradient the equation demands there; a solution is a curve that follows them.

THE DIRECTION FIELD

At each point, $\frac{dy}{dx}$ says what the gradient must be. Draw a short dash of that gradient at many points and the solution curves appear as the paths that follow the dashes. Two solution curves never cross — through each point runs exactly one.

Forming one

Most marks in this topic are for translating a sentence into symbols. The vocabulary is small and fixed.

ENGLISH	SYMBOLS
the rate of change of N with time	$\frac{dN}{dt}$
… is proportional to N	$\frac{dN}{dt} = kN$
… is inversely proportional to N	$\frac{dN}{dt} = \frac{k}{N}$
… decreases at a rate proportional to N	$\frac{dN}{dt} = -kN$
… proportional to the amount still to go	$\frac{dN}{dt} = k(A - N)$
… proportional to the product of N and $A - N$	$\frac{dN}{dt} = kN(A - N)$

PUT THE MINUS SIGN IN, AND THEN KEEP K POSITIVE

“Decreases at a rate proportional to N” is $\frac{dN}{dt} = -kN$ with $k > 0$. Writing $= kN$ and expecting k to come out negative works, but every later line then carries a sign you must remember. Declare it once.

Real cases. Radioactive decay $\frac{dN}{dt} = -kN$; Newton's law of cooling $\frac{d\theta}{dt} = -k(\theta - \theta_0)$; a population limited by resources $\frac{dP}{dt} = kP(M - P)$; a tank draining $\frac{dV}{dt} = -k\sqrt{h}$.

Separating the variables

An equation is **separable** if it can be written as a function of y times a function of x:

$$\frac{dy}{dx} = f(x)\,g(y) \Rightarrow \int \frac{dy}{g(y)} = \int f(x)\,dx$$

Every equation in this course is separable. The method is three lines:

1. get all the ys with dy and all the xs with dx;
2. integrate both sides, with **one** constant;
3. rearrange for y if the question asks for it.

The standard case. $\frac{dy}{dx} = ky$:

$$\int \frac{dy}{y} = \int k\,dx \Rightarrow \ln|y| = kx + c \Rightarrow y = Ae^{kx}$$

E^C BECOMES A NEW CONSTANT A

From $\ln y = kx + c$, exponentiating gives $y = e^{kx + c} = e^c e^{kx}$. Since e^c is just some positive constant, write it as A once and never carry c again. Allowing A to be negative covers $y < 0$ too.

ONE CONSTANT, AND ADD IT AT THE INTEGRATION

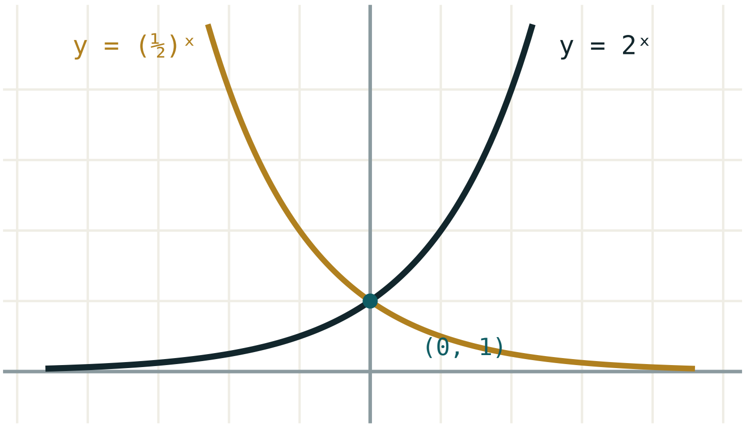
Adding c to both sides gives two constants where one will do. And adding it after rearranging — $y = Ae^{kx} + c$ — is simply a different, wrong, equation.

Particular solutions

An **initial condition** such as $y = 5$ when $x = 0$ fixes the constant. Substitute as soon as the general solution is written — before any further rearranging.

Worked case — decay. A radioactive sample obeys $\frac{dN}{dt} = -kN$, with $N = 200$ at $t = 0$ and a half-life of 5 days.

STEP	WORKING
separate	$\int \frac{dN}{N} = -\int k\,dt$
integrate	$\ln N = -kt + c$
general	$N = Ae^{-kt}$
$t = 0$	$A = 200$
half-life	$100 = 200e^{-5k} \Rightarrow k = \frac{\ln 2}{5} \approx 0.1386$
particular	$N = 200e^{-0.1386t}$



both pass through (0, 1); the x-axis is an asymptote

Exponential decay — the shape of every dN/dt = −kN.

Now any question is a substitution: after 12 days, $N = 200e^{-1.663} \approx 37.9$.

Interpreting

The last mark is usually for saying what the answer **means**.

MODEL	SOLUTION	LONG TERM
$\frac{dN}{dt} = kN$	$N = Ae^{kt}$	grows without limit — unrealistic eventually
$\frac{dN}{dt} = -kN$	$N = Ae^{-kt}$	$N \to 0$
$\frac{d\theta}{dt} = -k(\theta - \theta_0)$	$\theta = \theta_0 + Ae^{-kt}$	$\theta \to \theta_0$
$\frac{dP}{dt} = kP(M - P)$	logistic	$P \to M$

EQUILIBRIUM: SET THE RATE TO ZERO

Whatever makes $\frac{dy}{dx} = 0$ is an **equilibrium** value — the solution stays there for ever if it starts there, and usually approaches it otherwise. For cooling that is $\theta = \theta_0$, room temperature; for the logistic model it is $P = M$, the carrying capacity. You can read the long-term answer off the equation without solving it.

A caution about models. Unlimited exponential growth is never a description of reality for long; it is a description of the early stage, before whatever limits growth begins to act. Saying so is often worth a mark, and is always worth saying.

02 Worked examples

EXAMPLE 1 **Solve $\frac{dy}{dx} = \frac{x}{y}$ given that $y = 3$ when $x = 0$.**

1. $\int y\,dy = \int x\,dx$

   Separate.

2. $\frac{y^2}{2} = \frac{x^2}{2} + c$

3. $9 = 0 + 2c \Rightarrow c = 4.5$

   Substitute at once.

4. $y^2 = x^2 + 9$

ANSWER

$y = \sqrt{x^2 + 9}$ (taking the positive root)

EXAMPLE 2 **A sample decays with $\frac{dN}{dt} = -kN$. It starts at 200 and has a half-life of 5 days. Find N after 12 days.**

1. $N = Ae$^{$-kt$}, $A = 200$

2. $100 = 200e$^{$-5k$} $\Rightarrow e$^{$-5k$} $= 0.5$

3. $k = \frac{\ln 2}{5} \approx 0.1386$

4. $N = 200e$^{$-0.1386(12)$} ≈ 37.9

ANSWER

≈ 38 units

EXAMPLE 3

A body at $90°$ cools in a room at $20°$. After 10 minutes it is $60°$. Find its temperature after 25 minutes.

1. $\frac{d\theta}{dt} = -k(\theta - 20) \Rightarrow \theta = 20 + Ae^{-kt}$
2. $90 = 20 + A \Rightarrow A = 70$
3. $60 = 20 + 70e^{-10k} \Rightarrow e^{-10k} = \frac{4}{7} \Rightarrow k \approx 0.05596$
4. $\theta = 20 + 70e^{-1.399} \approx 37.3°$

ANSWER

$\approx 37.3°$

EXAMPLE 4

A population satisfies $\frac{dP}{dt} = 0.02P(500 - P)$. State the equilibrium values and the long-term population.

1. Set $\frac{dP}{dt} = 0$.
2. $P = 0$ or $P = 500$.
   Two equilibria.
3. For $0 < P < 500$ the rate is positive, so P rises.
4. $P \to 500$
   The carrying capacity.

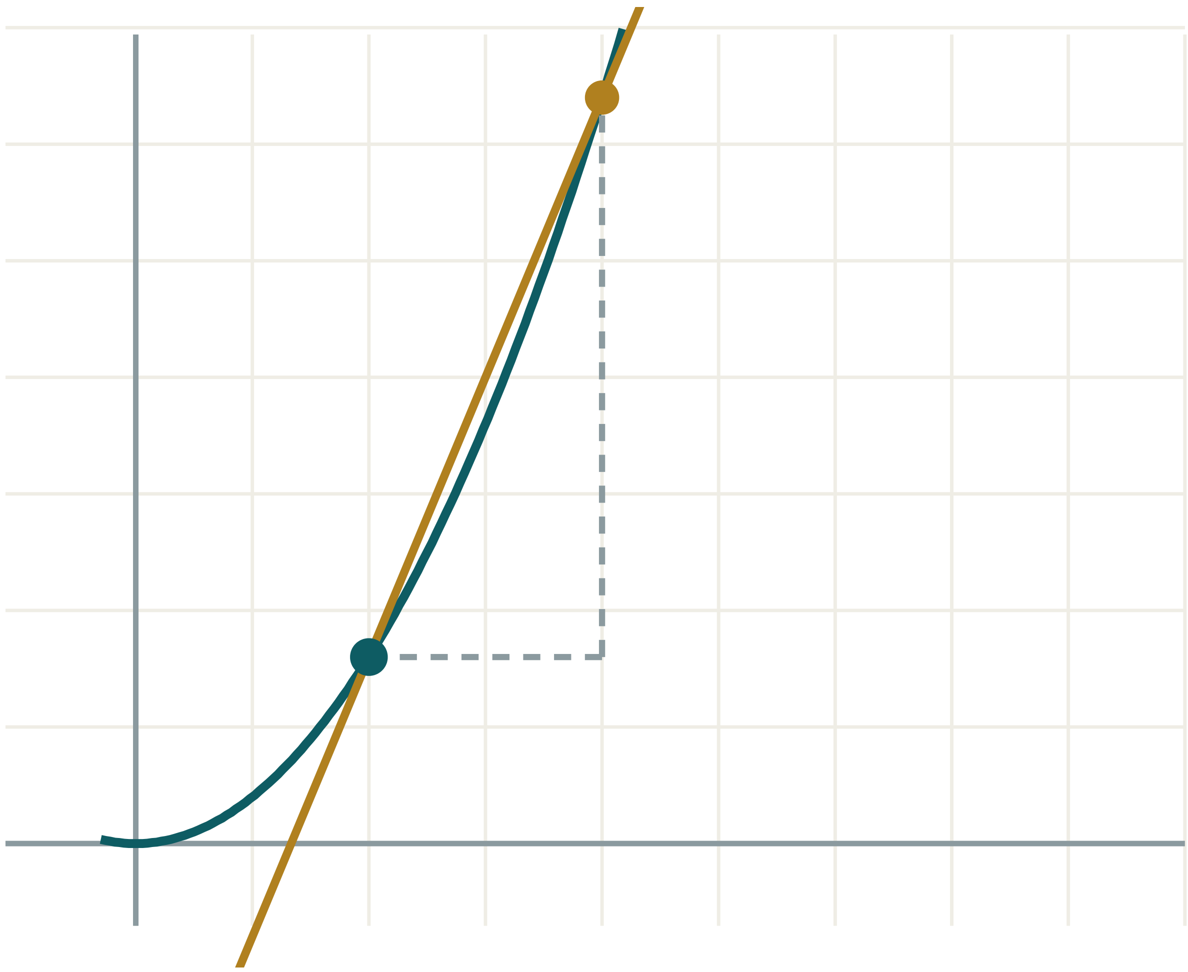
ANSWER

Equilibria 0 and 500; long term $P \to 500$

03 Interactive model

Cool a cup of hot water in front of the class, reading the thermometer every two minutes; the plotted points fall on the predicted curve.

A curve and its gradient



x **2** | h **2** | Δy / Δx **2.4** | f ′(x) exactly **1.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Differential equation	Differensial tenglama	Дифференциальное уравнение
Rate of change	O‘zgarish tezligi	Скорость изменения
Proportional	Proporsional	Пропорциональный
Separable equation	Ajraladigan tenglama	Уравнение с разделяющимися переменными
General solution	Umumiy yechim	Общее решение
Particular solution	Xususiy yechim	Частное решение
Initial condition	Boshlang‘ich shart	Начальное условие
Arbitrary constant	Ixtiyoriy o‘zgarmas	Произвольная постоянная
Exponential growth	Eksponensial o‘sish	Экспоненциальный рост
Exponential decay	Eksponensial kamayish	Экспоненциальное убывание
Direction field	Yo‘nalishlar maydoni	Поле направлений
Equilibrium value	Muvozanat qiymati	Равновесное значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The solution of a differential equation is:

- a number
- a function
- a gradient
- a constant

"Proportional to N" is:

- $\frac{dN}{dt} = kN$
- $\frac{dN}{dt} = \frac{k}{N}$
- $N = kt$
- $\frac{dN}{dt} = k$

A separable equation is solved by:

- differentiating
- separating and integrating
- squaring
- substituting

The general solution of $\frac{dy}{dx} = ky$:

- $y = kx + c$
- $y = Ae^{kx}$
- $y = \ln kx$
- $y = kx^2$

How many arbitrary constants does a first-order equation give?

- 0
- 1
- 2
- as many as you like

An equilibrium value is where:

- $y = 0$
- $\frac{dy}{dx} = 0$
- $x = 0$
- $k = 0$

For $\frac{d\theta}{dt} = -k(\theta - \theta_0)$, the long-term temperature is:

- 0
- θ_0
- ∞
- A

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $\frac{dy}{dx} = 2x$

   ANSWER $y = x^2 + c$

2. Solve $\frac{dy}{dx} = 3y$

   ANSWER $y = Ae^{3x}$

3. Solve $\frac{dy}{dx} = -2y$

   ANSWER $y = Ae^{-2x}$

4. Write “decreases at a rate proportional to N”

   ANSWER $\frac{dN}{dt} = -kN$

5. Write “rate proportional to the amount left, $A - N$”

   ANSWER $\frac{dN}{dt} = k(A - N)$

6. Equilibrium of $\frac{dy}{dt} = 3 - y$

   ANSWER $y = 3$

7. Number of constants in the general solution

   ANSWER 1

Medium · the standard question

7 PROBLEMS

1. Solve $\frac{dy}{dx} = \frac{x}{y}$ with $y(0) = 3$

   ANSWER $y = \sqrt{x^2 + 9}$

2. Solve $\frac{dy}{dx} = xy$ with $y(0) = 2$

   ANSWER $y = 2e^{x^2/2}$

3. Solve $\frac{dy}{dx} = \frac{y}{x}$ with $y(1) = 4$

   ANSWER $y = 4x$

4. Half-life 5 days, $N_0 = 200$: find k

   ANSWER ≈ 0.1386

5. Same: N after 12 days

   ANSWER ≈ 38

6. Cooling from $90°$ in a $20°$ room, $60°$ after 10 min: θ after 25 min

   ANSWER $\approx 37.3°$

7. Equilibria of $\frac{dP}{dt} = 0.02P(500 - P)$

   ANSWER 0 and 500

Hard · stretch and reason

7 PROBLEMS

1. Solve $\frac{dy}{dx} = y(1 - y)$ with $y(0) = \frac{1}{2}$

   ANSWER $y = \frac{1}{1 + e^{-x}}$

2. A tank drains with $\frac{dh}{dt} = -0.2\sqrt{h}$ from $h = 25$: time to empty

   ANSWER 50 s

3. A population doubles in 8 years under $\frac{dP}{dt} = kP$: find k and the time to triple

   ANSWER $k \approx 0.0866$; ≈ 12.7 years

4. Solve $\frac{dy}{dx} = \frac{1 + y^2}{1 + x^2}$ with $y(0) = 1$

   ANSWER $y = \tan(\arctan x + \frac{\pi}{4})$

5. A drug leaves the body with $\frac{dm}{dt} = -0.15m$ from 80 mg: time to fall below 10 mg

   ANSWER ≈ 13.9 h

6. Show that $y = \theta_0 + Ae^{-kt}$ solves Newton's law of cooling

   ANSWER $\frac{d\theta}{dt} = -kAe^{-kt} = -k(\theta - \theta_0)$

7. A savings account grows at 5% continuously: time to double

   ANSWER $\frac{\ln 2}{0.05} \approx 13.9$ years

07 Homework

Homework — 5 tasks

Every answer ends with one sentence saying what the solution means in the situation described.

1. Solve $\frac{dy}{dx} = \frac{2x}{y}$ given that $y = 4$ when $x = 0$.
2. A radioactive substance has a half-life of 8 days and starts at 500 units. Form and solve the differential equation, then find the amount after 20 days.
3. A cup of tea at $95°$ cools in a room at $22°$ and reaches $70°$ after 8 minutes. Find its temperature after 20 minutes.
4. A bacterial population satisfies $\frac{dP}{dt} = 0.4P$ with $P(0) = 1000$. Find the time to reach $10\ 000$, and say in one sentence why the model cannot hold for ever.
5. For $\frac{dy}{dt} = 6 - 2y$, state the equilibrium value and describe the behaviour of solutions starting above and below it.